

P08-R1 / 01 ZASILANIE CZUJNIKA I PRZEKAZNIK

R1 232k 1%: okolo 117 mA typ.; z rownan TI okolo 99..139 mA z tolerancja rezystora. Nie jest to limit 20 mA.

K1: COM 3/6, NO 4/5. Rozlacza +5V i powrot; AGND_SENSOR nie laczyć z GND poza K1.

U8 wymusza TPS_EN=0 przy brownout 3V3. R15/R16 zmniejszaja EN ponizej 0.66 V gdy 3V3 <1 V.

U8 zwalnia do 700 ms; przed pierwszym wlaczeniem odczekac 750 ms od SENSOR_OK=1. K1 moze zalaczyc wczesniej niz U1.

Switched current-limited sensor supply; review, hardware not tested

Prototype / schematic + PCB review

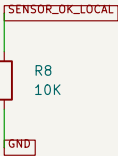
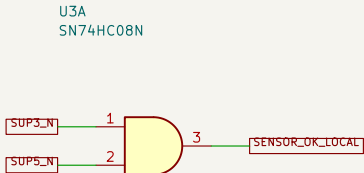
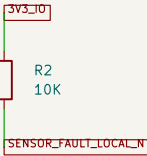
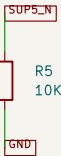
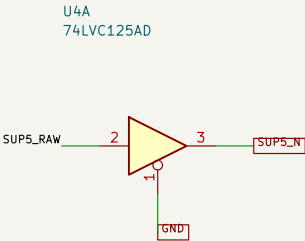
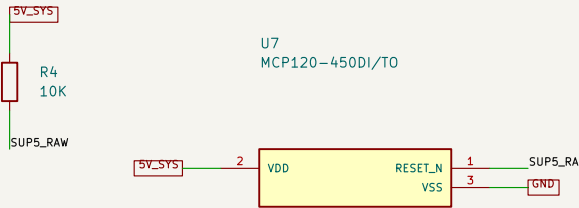
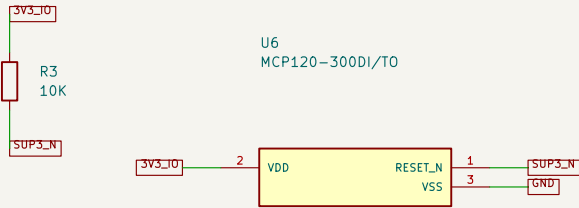
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Title: EGRLab P08 SENSOR

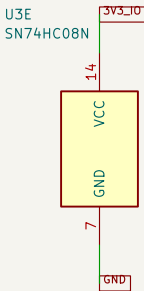
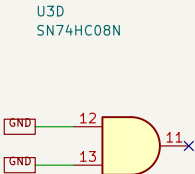
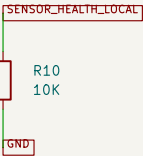
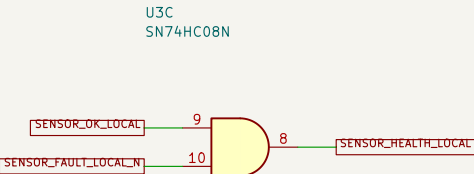
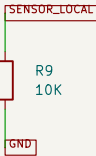
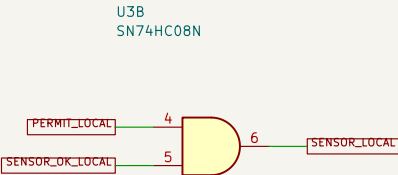
Size: A3	Date: 2026-09-27	Rev: P08-R1
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P08-R1 / 02 GOTOWOSC I WARUNKI ZEZWOLENIA

SENSOR_OK_LOCAL = SUP3_N & SUP5_N; niezależny od PERMIT i od napięcia za przekaźnikiem.



SENSOR_LOCAL = PERMIT_LOCAL & SENSOR_OK_LOCAL. HEALTH_LOCAL = SENSOR_OK_LOCAL & FAULT_N.



Brak sprzętowego latch FAULT. U1 ogranicza prąd; CORE musi zatrzasnąć błąd i cofnąć PERMIT.

Switched current-limited sensor supply; review, hardware not tested

Prototype / schematic + PCB review

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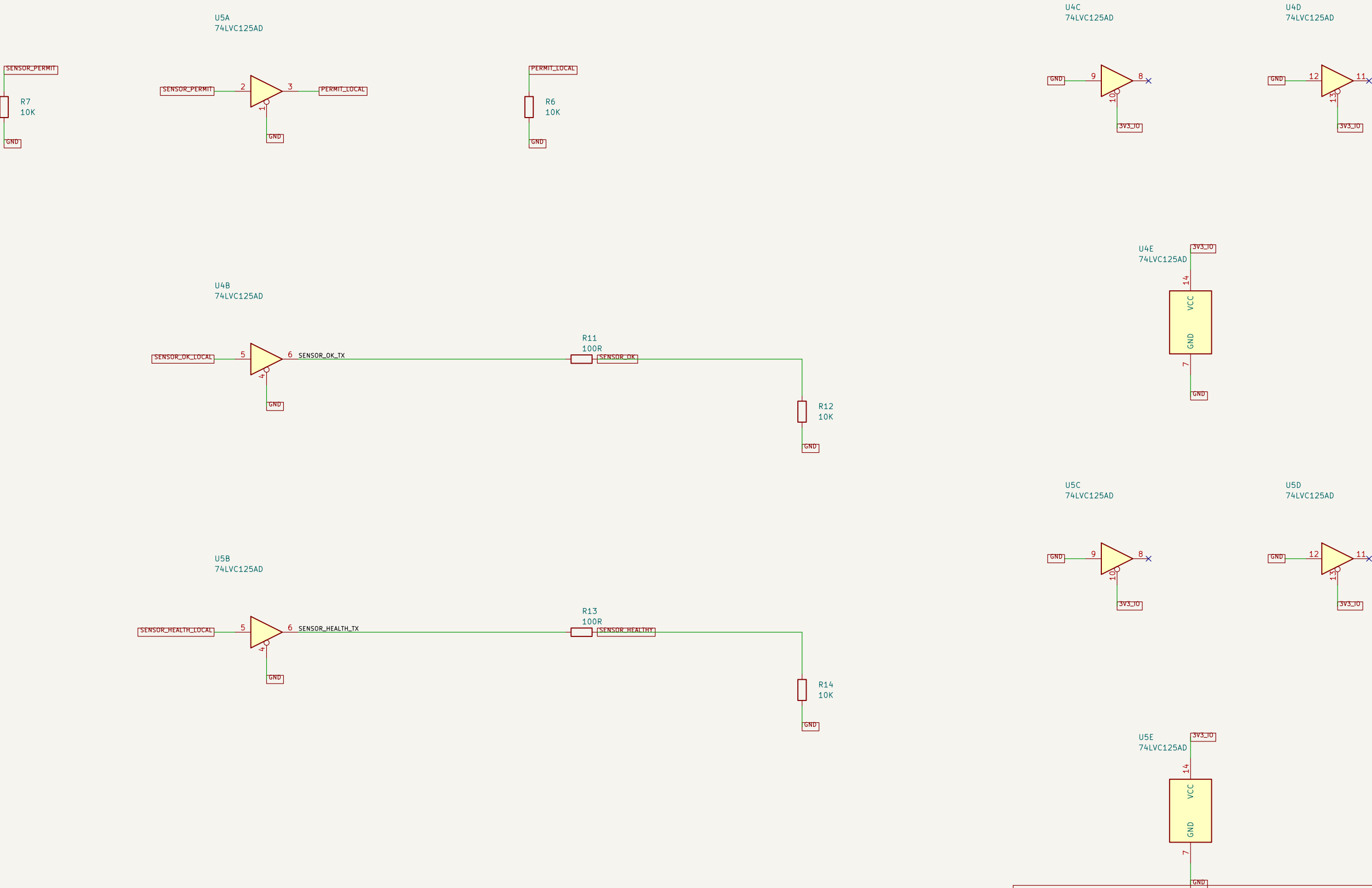
Title: EGRLab P08 SENSOR

Size: A3 Date: 2026-09-27
KiCad E.D.A. 10.0.6

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P08-R1 / 03 BUFORY SYGNALOW MIEDZY PLYTKAMI

U4/U5: Nexperia 74LVC125AD z loff, bez zamiany na HC125. Zasilanie obu 3V3_I0.



Wyjścia SENSOR_OK i SENSOR_HEALTHY: 100R szeregowo, 10k do GND. Brak zasilania -> LOW u odbiorcy.

HEALTHY=1 nie dowodzi zamknięcia K1, poprawnego mapowania pinów EGR ani obecności 5 V na czujniku.

Switched current-limited sensor supply; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /10/
File: 10.kicad_sch

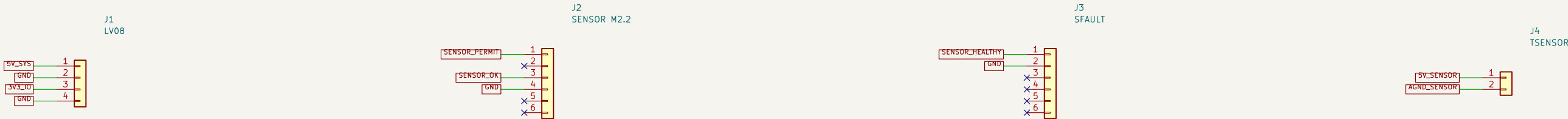
Title: EGRLab P08 SENSOR

Size: A3 Date: 2026-09-27
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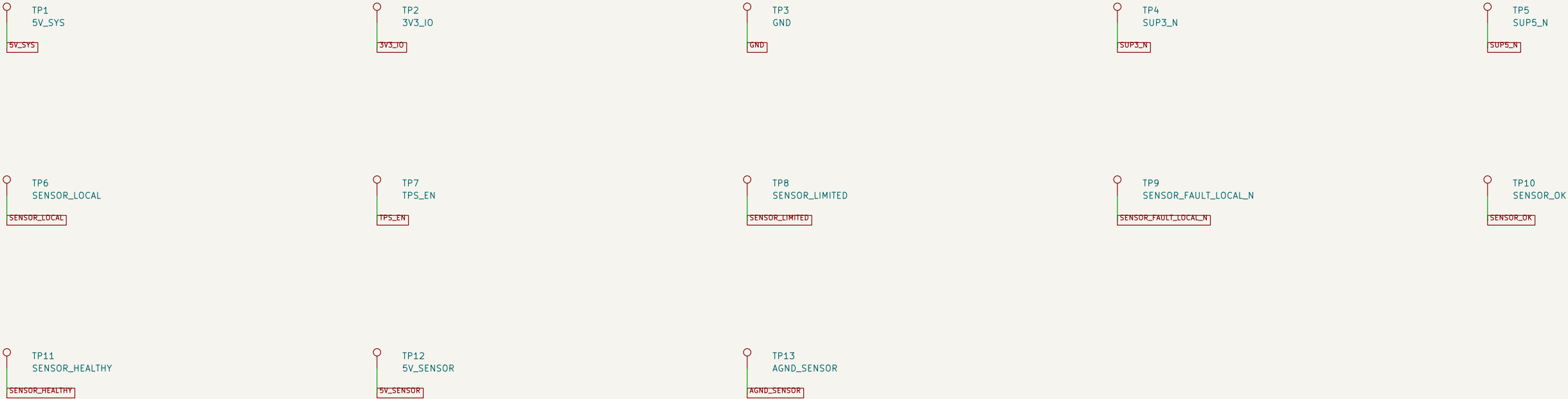
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P08–R1 / 04 WIAZKI I PUNKTY POMIAROWE

J1 -> P02/J8 (LV08). J2 -> P04–R2.1/J4 (SENSOR 6p, KEY2). J3 -> P03–R2/J6 (SFAULT 6p, KEY3).



J1/J2/J3: lutowane PTH, kotwa 12 mm. J4: gniazdo meskie na PCB; wiazka lutowana po stronie przyszlej P11.



TEST: mapowanie 5V/GND czujnika zatwierdzone poza P08. LOGGER: SENSOR_PERMIT=0; zasilanie czujnika pochodzi z ECU.

Switched current–limited sensor supply; review, hardware not tested
Prototype / schematic + PCB review
Sheet: /CONNECT/
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Title: EGRLab P08 SENSOR		
Size: A3	Date: 2026–09–27	Rev: P08–R1
KiCad E.D.A. 10.0.6		Id: 4/4